

REST API for External Guest Authentication

External Guest Authentication

Zone Settings

General

Guest Services

Wireless

Radius Server

GUEST SERVICES

Enable Guest Service

☒

Enable Inter-guest Communication

☐

Enable External Guest Authentication

☒

Configure

Enable Captive Portal Authentication

☐

Configure

Enable Policy Page without authentication

☐

Configure

Custom Authentication Page

☐

Configure

Enable Post Authentication Page

☐

Post Authentication Page

Bypass Guest Authentication

☐

All MAC Addresses

Redirect SMTP traffic to

☐

X0 IP

Deny Networks

☐

X0 IP

Pass Networks

☐

X0 IP

Max Guests

10

This feature supports both cgi and rest API since Gen6.

The following cgi APIs have been deprecated since 7.3.2.

login	externalGuestLogin.cgi
logout	externalGuestLogoff.cgi
create guest account	createGuestAccount.cgi
update guest session	externalGuestUpdateSession.cgi

Currently, the rest API `/lhmapi/externalAAAGuest` replace of those cgi APIs.

1.1 Login

The Figure 1 shows the user login process.

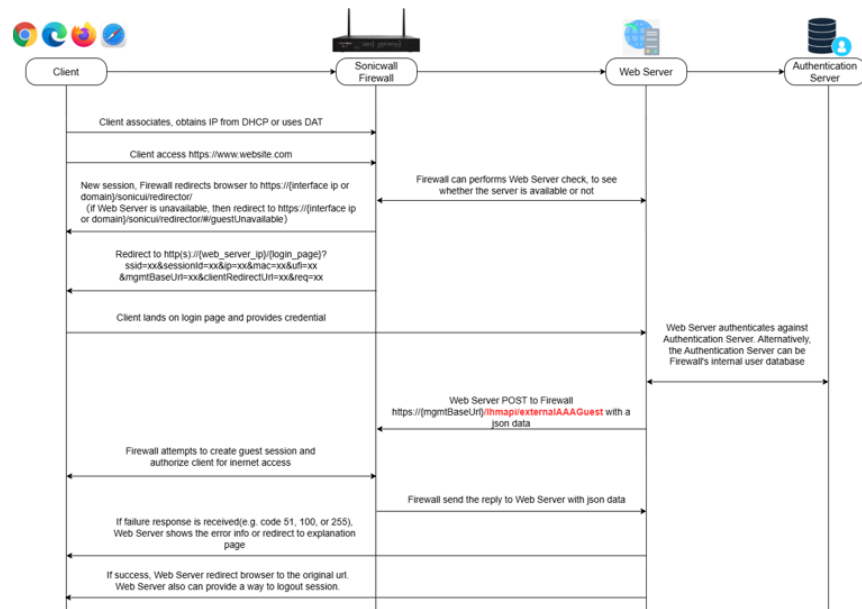


Figure 1. User Login Flow

The meaning of parameters in redirect url.

```

1 http(s)://{web_server_ip}/{login_page}?ssid=xx&sessionId=xx&ip=xx&
2 mac=xx&ufi=xx&mgmtBaseUrl=xx&clientRedirectUrl=xx&req=xx&hmac=xx
  
```

ssid	The ESSID (wireless network name) of the wireless network to which the redirected client was associated
sessionId	A 32 byte hex representation of a 16 byte MD5 hash value generated by the SonicWALL, which will be used by the SonicWALL and the LHM Server for indexing clients (e.g. "11aa3e2f5da3e12ef978ba120d2300ff")
ip	The client IP address
mac	The client MAC address
ufi	The SonicWALL Unique Firewall Identifier. To be used for site identification, if desired
mgmtBaseUrl	The protocol, IP address, and port on the Firewall with which the IP will subsequently communicate between Web Server and Firewall
clientRedirectUrl	The protocol, IP address (and optionally port) on the Firewall that the Web Server will use for client redirection
req	The client's originally requested URL, if any, URL encoded

hmac	If firewall enable the Message Authentication, this value is generated from Firewall
------	--

During the login process, the Web Server sends a POST to <https://{mgmtBaseUrl}/lhmapl/externalAAAAGuest> with a json formatted data just like below:

Request Body

```

1 POST /lhmapl/externalAAAAGuest
2 Content-Type: application/json
3 {
4   "info": {
5     "action": 1,
6     "sessId": "xxx",
7     "userName": "xxx",
8     "passwd": "xxx",
9     "sessionLifetime": "xxx",
10    "idleTimeout": "xxx",
11    "hmac": "xxx",
12    "maxRx": "xxx",
13    "maxTx": "xxx",
14    "quotaCycleType": "xxx",
15    "cycleSessionLifeTime": "xxx",
16    "cycleMaxRx": "xxx",
17    "cycleMaxTx": "xxx"
18  }
19 }
```

Parameters

Field	Required	Description
action	Yes	Action type. 1 indicates login.
sessId	Yes	A 32 byte hex representation of a 16 byte MD5 hash value generated by the SonicWALL, which will be used by the SonicWALL and the LHM Server for indexing clients.
userName	Yes	Username for authentication.
passwd	Optional	User password. If present, the firewall will verify credentials using the local user database. Otherwise, LHM server itself should do the authentication, for exapmle, LDAP, Radius, etc.
sessionLifetime	Yes	Session lifetime in seconds, the value is greater than or equal to 0. 0 means unlimited.
idleTimeout	Yes	Idle timeout in seconds. Minimum value is 60 seconds. Must be less than sessionLifetime.
hmac	Conditional	Required if HMAC authentication is enabled on the firewall.

		HMAC supports the following algorithms: MD5, SHA-1, and SHA-256.
maxRx	Yes	Maximum RX traffic in MB, the value is greater than or equal to 0. 0 means unlimited.
maxTx	Yes	Maximum TX traffic in MB, the value is greater than or equal to 0. 0 means unlimited.
quotaCycleType	Yes	Quota cycle type: 0 = No quota cycle 1 = Per day 2 = Per week 3 = Per month
cycleSessionLifetime	Yes	Cycle session lifetime in seconds, the value is greater than or equal to 0. 0 means unlimited.
cycleMaxRx	Yes	Cycle RX quota in MB, the value is greater than or equal to 0. 0 means unlimited.
cycleMaxTx	Yes	Cycle TX quota in MB, the value is greater than or equal to 0. 0 means unlimited.

HMAC Calculation

If Message Authentication is enabled on the firewall, the hmac field becomes required in the post body.

For the login process,

- If the query has hmac, then the login page should verify the HMAC firstly.

The req is encoded by sonicwall, so before calculate the HMAC, req should be encode firstly.

Then the hmac calculated on LHM server side should be compared with the hmac in query, if doesn't match, then refuse to login.

```

1 $reqEncoded = $req;
2 $reqEncoded = str_replace("%", "%25", $reqEncoded);
3 $reqEncoded = str_replace(":", "%3A", $reqEncoded);
4 $reqEncoded = str_replace(" ", "%20", $reqEncoded);
5 $reqEncoded = str_replace("?", "%3F", $reqEncoded);
6 $reqEncoded = str_replace("+", "%2B", $reqEncoded);
7 $reqEncoded = str_replace("&", "%26", $reqEncoded);
8 $reqEncoded = str_replace("=", "%3D", $reqEncoded);

```

The HMAC is calculated by concatenating the following fields in order, the params are got from the query.

```

1 ssid
2 sessionId
3 ip
4 mac
5 ufi
6 mgmtBaseUrl
7 clientRedirectUrl

```

- When click the login button, the hmac should be included in post body. HMAC is calculated by concatenating the following fields in order:

```

1 sessId
2 urlencode(userName)
3 sessionLifetime
4 idleTimeout
5 maxRx
6 maxTx
7 quotaCycleType
8 cycleSessionLifeTime
9 cycleMaxRx
10 cycleMaxTx

```

Notes:

- Fields are concatenated in the exact order listed above
- username needs to encode
- Fields are concatenated without any delimiter
- Fields are concatenated without spaces
- The hmac field itself is not included in the calculation
- All values are used as string format before calculation
- If HMAC is enabled on the firewall, hmac becomes mandatory

Quota Cycle Behavior

If quotaCycleType > 0, the backend will use the following parameters:

- cycleSessionLifeTime
- cycleMaxRx
- cycleMaxTx

In this case, cycleSessionLifeTime must not exceed the cycle duration:

quotaCycleType	Cycle Duration	Max cycleSessionLifeTime
1 (Per day)	24 hours	24 * 3600
2 (Per week)	7 days	7 * 24 * 3600
3 (Per month)	30 days	30 * 24 * 3600

If quotaCycleType = 0, the backend will use:

- sessionLifetime
- maxRx
- maxTx

Constraints

- All numeric values are passed as string type
- idleTimeout ≥ 60
- idleTimeout < sessionLifetime

- sessionLifetime ≥ 0
- cycleSessionLifeTime ≤ cycle duration
- 0 means Unlimited for all quota-related values

Example

```
1 {
2   "info": {
3     "action": 1,
4     "sessId": "329d6adf434f26d67c4368e74ed42f69",
5     "userName": "testUser",
6     "sessionLifetime": "3600",
7     "idleTimeout": "600",
8     "maxRx": "1024",
9     "maxTx": "1024",
10    "quotaCycleType": "2",
11    "cycleSessionLifeTime": "86400",
12    "cycleMaxRx": "2048",
13    "cycleMaxTx": "2048"
14  }
15 }
```

SonicOS 7.3.2 Limitation

The `lhmapr/externalAAAGuest` has some limitation, we will remove those limitation later.

Post Body	Limitation
<pre>{ "info": { "action": 1, "sessId": "xxx", "userName": "xxx", "passwd": "xxx", "sessionLifetime": "xxx", "idleTimeout": "xxx", "hmac": "xxx", "maxRx": "xxx", "maxTx": "xxx", "quotaCycleType": "xxx", "cycleSessionLifeTime": "xxx", "cycleMaxRx": "xxx", "cycleMaxTx": "xxx" } }</pre>	In SonicOS 7.3.2, the value range that the backend API can accept is: 0 < sessionLifetime <= 9999 0 < idleTimeout <= sessionLifetime 0 <= cycleSessionLifeTime <= 9999

1.2 Logout

The Figure 2 shows the user logout process.

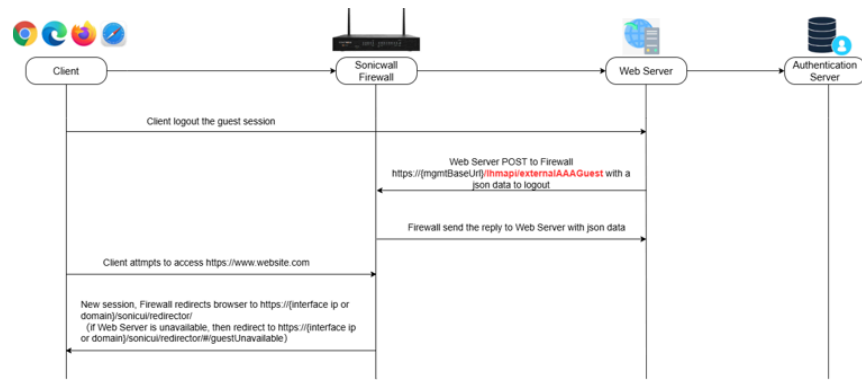


Figure 2. User Logout Flow

During the logout process, the Web Server sends a POST to `https://{ mgmtBaseUrl}/lhmap/externalAAAAGuest` with a json formatted data just like below:

Request Body

```

1 POST /lhmap/externalAAAAGuest
2 Content-Type: application/json
3 {
4   "info": {
5     "action": 2,
6     "sessId": "xxx",
7     "eventId": "xxx",
8     "hmac": "xxx"
9   }
10 }

```

Parameters

Field	Required	Description
action	Yes	Action type. 2 indicates session logout.
sessId	Yes	A 32 byte hex representation of a 16 byte MD5 hash value generated by the SonicWALL, which will be used by the SonicWALL and the LHM Server for indexing clients.
eventId	Yes	Logout event reason. See Event ID Table below.
hmac	Conditional	Required if Message Authentication (HMAC) is enabled on the firewall. HMAC supports the following algorithms: MD5, SHA-1, and SHA-256.

Event ID Values

Value	Description
1	Logout initiated by guest user
2	Logout initiated by administrator
3	Logout due to session timeout
4	Logout due to idle timeout

HMAC Calculation

If Message Authentication is enabled on the firewall, the hmac field becomes required.

HMAC is calculated by concatenating the following fields in order:

1	sessId
2	eventId

Notes

- Fields are concatenated in the exact order listed above
- Fields are concatenated without any delimiter
- Fields are concatenated without spaces
- The hmac field itself is not included in the calculation
- All values are used as string format before calculation
- If HMAC is enabled on the firewall, hmac becomes mandatory

Constraints

- All values are passed as string type
- eventId must be one of: 1, 2, 3, 4, 5
- If HMAC is enabled, request without hmac will be rejected

Example

```

1 {
2   "info": {
3     "action": 2,
4     "sessId": "329d6adf434f26d67c4368e74ed42f69",
5     "eventId": "1"
6   }
7 }
```

1.3 Create guest account

The Figure 3 shows the user account creation process.

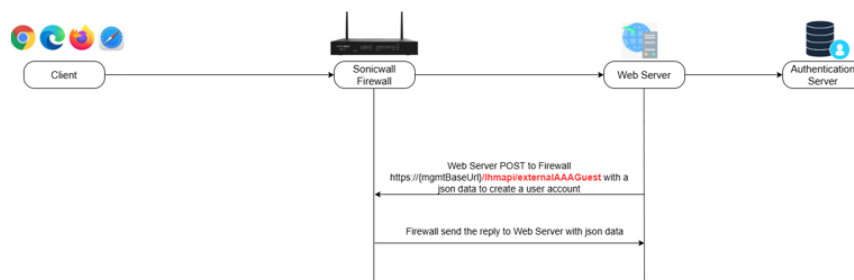


Figure 3. User Account Creation Flow

During the creation of guest account process, the Web Server sends a POST to `https://{mgmtBaseUrl}/lhmapi/externalAAAGuest` with a json formatted data just like below:

Request Body

```

1 POST /lhmapi/externalAAAGuest
2 Content-Type: application/json
3 {
4   "info": {
```



```

5      "action": 3,
6      "userName": "xxx",
7      "passwd": "xxx",
8      "comment": "xxx",
9      "hmac": "xxx",
10     "accountLifetime": "xxx",
11     "sessionLifetime": "xxx",
12     "idleTimeout": "xxx",
13     "uniqueLogin": "xxx",
14     "activateNow": "xxx",
15     "autoPrune": "xxx",
16     "maxRx": "xxx",
17     "maxTx": "xxx",
18     "quotaCycleType": "xxx",
19     "cycleSessionLifeTime": "xxx",
20     "cycleMaxRx": "xxx",
21     "cycleMaxTx": "xxx"
22   }
23 }

```

Parameters

Field	Required	Description
action	Yes	Action type. 3 indicates account creation.
userName	Yes	Username of the account.
passwd	Yes	Password of the account.
comment	Yes	Account description.
hmac	Conditional	Required if Message Authentication (HMAC) is enabled on the firewall. HMAC supports the following algorithms: MD5, SHA-1, and SHA-256.
accountLifetime	Yes	Account lifetime in seconds. Valid range: (0, 31536000] (up to 365 days).
sessionLifetime	Yes	Session lifetime in seconds. Valid range: [0, accountLifetime]. 0 means unlimited.
idleTimeout	Yes	Idle timeout in seconds. Minimum value is 60 seconds. Must be less than sessionLifetime.
uniqueLogin	Yes	Allow only one active login: 0 = Disable 1 = Enable
activateNow	Yes	Activate account immediately: 0 = Disable, activate account upon first login 1 = Enable, activate account immediately
autoPrune	Yes	Automatically remove expired account: 0 = Disable 1 = Enable

maxRx	Yes	Maximum RX quota in MB. 0 means unlimited.
maxTx	Yes	Maximum TX quota in MB. 0 means unlimited.
quotaCycleType	Yes	Quota cycle type: 0 = No quota cycle 1 = Per day 2 = Per week 3 = Per month
cycleSessionLifeTime	Yes	Cycle session lifetime in seconds. 0 means unlimited.
cycleMaxRx	Yes	Cycle RX quota in MB. 0 means unlimited.
cycleMaxTx	Yes	Cycle TX quota in MB. 0 means unlimited.

Quota Cycle Behavior

If quotaCycleType > 0, the backend will use the following parameters:

- cycleSessionLifeTime
- cycleMaxRx
- cycleMaxTx

In this case, cycleSessionLifeTime must not exceed the cycle duration:

quotaCycleType	Cycle Duration	Max cycleSessionLifeTime
1 (Per day)	24 hours	24 * 3600
2 (Per week)	7 days	7 * 24 * 3600
3 (Per month)	30 days	30 * 24 * 3600

If quotaCycleType = 0, the backend will use:

- sessionLifetime
- maxRx
- maxTx

HMAC Calculation

If Message Authentication is enabled on the firewall, the hmac field becomes required.

HMAC is calculated by concatenating the following fields in order:

```

1  userName
2  passwd
3  comment
4  accountLifetime
5  sessionLifetime
6  idleTimeout

```

```

7 uniqueLogin
8 activateNow
9 autoPrune
10 maxRx
11 maxTx
12 quotaCycleType
13 cycleSessionLifeTime
14 cycleMaxRx
15 cycleMaxTx

```

Notes:

- Fields are concatenated in the exact order listed above
- Fields are concatenated without any delimiter
- Fields are concatenated without spaces
- The hmac field itself is not included in the calculation
- All values are used as string format before calculation
- If HMAC is enabled on the firewall, hmac becomes mandatory

Constraints

- All numeric values are passed as string type
- $\text{accountLifetime} \leq 31536000$
- $0 \leq \text{sessionLifetime} \leq \text{accountLifetime}$
- $\text{idleTimeout} \geq 60$
- $\text{idleTimeout} < \text{sessionLifetime}$
- $\text{cycleSessionLifeTime} \leq \text{cycle duration}$
- 0 means Unlimited for all quota-related values

Example

```

1 {
2   "info": {
3     "action": 3,
4     "userName": "guest001",
5     "passwd": "password",
6     "comment": "Temporary guest account",
7     "accountLifetime": "86400",
8     "sessionLifetime": "3600",
9     "idleTimeout": "600",
10    "uniqueLogin": "1",
11    "activateNow": "1",
12    "autoPrune": "1",
13    "maxRx": "1024",
14    "maxTx": "1024",
15    "quotaCycleType": "1",
16    "cycleSessionLifeTime": "86400",
17    "cycleMaxRx": "2048",
18    "cycleMaxTx": "2048"
19  }
20 }

```

SonicOS 7.3.2 Limitation

The `lhmapr/externalAAAGuest` has some limitation, we will remove those limitation and add the HMAC validation later.

Post Body	Limitation
<pre>{ "info": { "action": 3, "userName": "xxx",</pre>	1. In SonicOS 7.3.2, the value range that the backend API can accept is:

```

"passwd": "xxx",
"comment": "xxx",
"hmac": "xxx",
"accountLifetime": "xxx",
"sessionLifetime": "xxx",
"idleTimeout": "xxx",
"uniqueLogin": "xxx",
"activateNow": "xxx",
"autoPrune": "xxx",
"maxRx": "xxx",
"maxTx": "xxx",
"quotaCycleType": "xxx",
"cycleSessionLifeTime": "xxx",
"cycleMaxRx": "xxx",
"cycleMaxTx": "xxx"
}
}

```

0 < accountLifetime <= 9999
0 < sessionLifetime <= accountLifetime
0 < idleTimeout <= sessionLifetime
0 <= cycleSessionLifeTime <= 9999

2. Please don't enable **Message Authentication** when using this API to create guest account.

Zone Settings

General Guest Services Wireless Radius Server

GO BACK

General Auth Pages Web Content Advanced

Web Server 2 Port 443
Connection Timeout 15

MESSAGE AUTHENTICATION

Enable Message Authentication ☐

Authentication Method HMAC-SHA256
Shared Secret *****
Confirm Shared Secret *****

1.4 Update guest session

The Figure 4 shows the user update session process.

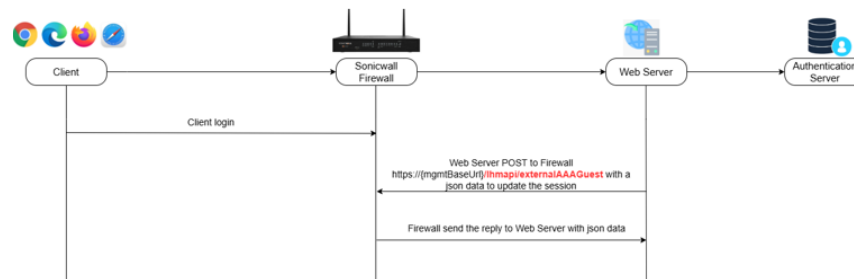


Figure 4. User Update Session Flow

During the update of session process, the WS sends a POST to `https://{ mgmtBaseUrl}/lhmapirexternalAAAGuest` with a json formatted data just like below:

Request Body

```

1 POST /lhmapirexternalAAAGuest
2 Content-Type: application/json
3 {
4   "info": {
5     "action": 4,
6     "sessId": "xxx",
7     "userName": "xxx",
8     "sessionLifetime": "xxx",
9     "idleTimeout": "xxx",
10    "hmac": "xxx",
11    "maxTx": "xxx",
12    "maxRx": "xxx",
13    "quotaCycleType": "xxx",
14    "cycleSessionLifeTime": "xxx",
15    "cycleMaxTx": "xxx",
16    "cycleMaxRx": "xxx"
17  }
18 }

```

Parameters

Field	Required	Description
action	Yes	Action type. 4 indicates session update.
sessId	Optional	Session ID. If provided, only the specified session will be updated. Otherwise, all sessions logged in with this userName will be updated.
userName	Yes	Username of the session
sessionLifetime	Yes	Session lifetime in seconds.
idleTimeout	Yes	Idle timeout in seconds. Minimum value is 60 seconds and must be less than sessionLifetime.
hmac	Conditional	Required if Message Authentication (HMAC) is enabled on the firewall. HMAC supports the following algorithms: MD5, SHA-1, and SHA-256.
maxTx	Yes	Maximum TX in MB. 0 means unlimited.
maxRx	Yes	Maximum RX in MB. 0 means unlimited.
quotaCycleType	Yes	Quota cycle type: 0 = No quota cycle 1 = Per day 2 = Per week 3 = Per month
cycleSessionLifeTime	Yes	Cycle session lifetime in seconds. 0 means unlimited.
cycleMaxTx	Yes	Cycle TX quota in MB. 0 means unlimited.
cycleMaxRx	Yes	Cycle RX quota in MB. 0 means unlimited.

Notes

- If sessId is present, only the specified session will be updated.
- If sessId is not present, all sessions associated with userName will be updated.
- When quotaCycleType = 0, cycle-related parameters may be ignored.
- All numeric values are provided as string type.
- 0 means Unlimited for quota-related parameters.

Quota Cycle Behavior

If quotaCycleType > 0, the backend will use the following parameters:

- cycleSessionLifeTime
- cycleMaxRx
- cycleMaxTx

In this case, cycleSessionLifeTime must not exceed the cycle duration:

quotaCycleType	Cycle Duration	Max cycleSessionLifeTime
1 (Per day)	24 hours	24 * 3600
2 (Per week)	7 days	7 * 24 * 3600
3 (Per month)	30 days	30 * 24 * 3600

If quotaCycleType = 0, the backend will use:

- sessionLifetime
- maxRx
- maxTx

HMAC Calculation

If firewall enables Message Authentication, the hmac field is required.

If request contains sessId, HMAC is calculated by concatenating the following fields in order:

```
1 sessId
2 userName
3 sessionLifetime
4 idleTimeout
5 maxRx
6 maxTx
```

Otherwise, HMAC is calculated by concatenating the following fields in order:

```
1 userName
2 sessionLifetime
3 idleTimeout
4 maxRx
5 maxTx
```

Notes

- Fields are concatenated in the exact order listed above
- Fields are concatenated without any delimiter
- Fields are concatenated without spaces
- The hmac field itself is not included in the calculation
- All values are used as string format before calculation
- If HMAC is enabled on the firewall, hmac becomes mandatory

Example

```
1 {
2   "info": {
3     "action": 4,
4     "sessId": "329d6adf434f26d67c4368e74ed42f69",
5     "userName": "guest01",
6     "sessionLifetime": "3600",
7     "idleTimeout": "600",
```

```

8      "maxTx": "1024",
9      "maxRx": "1024",
10     "quotaCycleType": "0",
11     "cycleSessionLifeTime": "0",
12     "cycleMaxTx": "0",
13     "cycleMaxRx": "0"
14   }
15 }

```

SonicOS 7.3.2 Limitation

The `lhmapl/externalAAAGuest` has some limitation, we will remove those limitation later.

Post Body	Limitation
<pre> { "info": { "action": 4, "sessId": "xxx", "userName": "xxx", "sessionLifetime": "xxx", "idleTimeout": "xxx", "hmac": "xxx", "maxTx": "xxx", "maxRx": "xxx", "quotaCycleType": "xxx", "cycleSessionLifeTime": "xxx", "cycleMaxTx": "xxx", "cycleMaxRx": "xxx" } } </pre>	In SonicOS 7.3.2, the value range that the backend API can accept is: 0 < sessionLifetime <= 9999 0 < idleTimeout <= sessionLifetime 0 <= cycleSessionLifeTime <= 9999

1.5 API Response

The SonicWALL returns a json response in the following format:

```

1 {
2   "code": "xxx",
3   "message": "xxx"
4 }

```

If logout is successfully, then the response is

```

1 {
2   "RxBytes": "xx",
3   "TxBytes": "xx"
4 }

```

The {response code} includes one of the values listed in the following table:

Response Code	Response Meaning
50	Login succeeded
51	Session limit exceeded
100	Login failed -- access reject

251	Msg. Auth failed -- Invalid HMAC
253	Invalid session ID
254	Invalid or missing parameter
255	Internal error
256	Invalid account lifetime, the valid value is greater than zero and less than or equal to 365 days
257	Invalid (cycle)session, the valid value is greater than or equal to zero and less than account lifetime
258	Invalid idle timeout, the valid value is greater than or equal to one minute and less than session lifetime
259	The cycle session lifetime can not be greater than the cycle time